High polyphenol, low probiotic diet for weight loss because of intestinal microbiota interaction.

The relative proportion of Bacteroidetes to Firmicutes is decreased in obese people. This imbalance in gut microbiota generates signals controlling the expression of genes by the epithelial intestinal cells. Both dairy and non-dairy probiotics increase body weight, reportedly through Lactobacillus species growth in the gut. On the other hand, daily intake of some fruits and drinks such as three apples or three pears or grapefruit, or green tea, which all are rich in polyphenols, can significantly reduce body weight in obese people. Metabolism of polyphenols by microbiota involves the cleavage of glycosidic linkages. Glycans, which are the product of glycosidic cleavage, are necessary for survival of the intestinal microbiota as a nutrient foundation. There are two pivotal points: (i) Firmicutes possess a disproportionately smaller number of glycan-degrading enzymes than Bacteroidetes, (ii) Firmicutes are more repressed than the Bacteroidetes by phenolic compounds' antimicrobial properties. The Bacteroidetes community prevails following dietary polyphenol intake and its fermentation to phenolic compounds, due to having more glycan-degrading enzymes, so this may thus be a mechanism by which dietary polyphenols exert their weight lowering effect. I suggest that future studies utilize clone libraries and fingerprinting techniques enabling identification of the composition and community structure of the microbiota, and dot blot hybridization or fluorescent in situ hybridization to analyze abundance of particular taxa in obese and individuals. A supplementation with polyphenols with high bioavailability in obese individuals with higher Firmicutes/Bacteroides community ratio phenotype, when associated to a probiotic restricted diet, is proposed for weight loss; this hypothesis could have relevant implication in planning a successful dietary regimen and/or neutraceutical/pharmaceutical preparations for achieving and maintaining a normal body weight in obese individuals, especially including much more use of polyphenol-rich foodstuffs and/or polyphenol-rich syrups, and including low amounts of probiotic-rich foodstuffs like yogurt, soy yogurt, or as probiotic supplements.